Noah Hart

Gameplay Programmer | Technical Designer

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Summary

Gameplay programmer and technical designer with 6+ years of coding experience and graduate training in the University of Utah's Master of Entertainment Arts & Engineering program, Engineering Track. As Lead Engineer on No Rail, I helped a 15-person team ship an Unreal Engine game on Steam. I build modular C++ gameplay, AI, and designer-friendly tools. I help teams iterate and deliver reliable features through debugging, clear documentation, and flexible systems.

Education

University of Utah 2024–2026
Master of Entertainment Arts & Engineering — Engineering Track

- Developed gameplay systems, AI, and prototypes in a multidisciplinary studio with production milestones.
- Applied sprint planning, playtesting, documentation, and system integration to team projects, including No Rail on Steam.
- **Recognition:** Rockin Derby — IndieCade selection and nomination.

University of California, Santa Cruz 2019–2023
B.S. Computer Science | GPA: 3.8

- Coursework: Data Structures, Algorithms, Computer Graphics, AI, and Game Design & Development.

Technical Skills & Interests

Languages: C++, C#, C, Python, Lua, JavaScript, HTML; Unreal Blueprints
Engines & Tools: Unreal Engine 5, Unity, openFrameworks, Git, Perforce, Visual Studio, Blender, Maya, Jira, Trello
Gameplay & Architecture: GAS, AI (behavior trees, pathfinding, GOAP), state machines, procedural content generation (PCG), object pooling; Programming patterns (State, Command, Type Object)
Mathematics: Linear algebra (vectors, matrices, coordinate transforms, dot/cross products), calculus, trigonometry
Development: Debugging, profiling, prototyping, mobile development, problem solving, documentation, team leadership, collaboration, sprint-based production
Interests: Piloting (PPL training), piano, volleyball (playing and coaching), gaming, graphics development

Selected Projects

No Rail 2025–2026
Lead Engineer, Systems & Level Designer | UE5 / C++ | Team of 15 | Released on Steam

- Built modular train generation with streamed levels, random car selection, and level-scoped spawners.
- Reworked encounters around terminal activation; integrated survival, capture-zone, and defense objectives with shared progression and a game-loop tutorial.
- Built station timing and return progression; led engineering sprints, debugging, and design/art integration.

C++ Gameplay Systems & Combat Design 2025–2026

- Developed modular UE5 combat with GAS, health/damage, and stamina systems, separating reusable gameplay logic from individual ability behavior.
- Built CineCaster, a multiplayer cinematic camera plugin supporting dynamic player switching.
- Implemented seeded Perlin-noise terrain and constrained object placement in Pet Fish Simulator, using the Strategy pattern to support configurable scatter and cluster placement.

AI Systems & Pathfinding 2026

- Implemented steering, A*, Dijkstra, behavior trees, and goal-oriented action planning; compared pathfinding heuristics using runtime and search metrics.
- Combined A* navigation, ghost behavior trees, and procedural maze generation in playable Pac-Man; deployed interactive browser demos using WebAssembly.

Rockin Derby (Alt Control) & Rapid Prototyping 2024–2025
Hardware & UI Engineer | Unity / C#

- Built three Unity/Unreal prototypes in 2–3 weeks each, including mobile games, under themed constraints, serious games, and retro remixes.
- Built ESP32/Arduino controllers for Alt Control, integrating motion input with Unity for a rocking-horse racer.
- Processed accelerometer/gyroscope input with calibration, dead zones, and smoothing for the rocking-horse kart racer showcased at the university launch event.

Current Projects: Project After Hours; Squirrel Life
Previous Projects: Graphics Engine (Flappy Squirrel); Castle of Cardboard (Mobile); Weapon Wheel & Gadget System

Experience

Teaching Assistant — University of Utah 2025–2026

- As Head TA, mentored capstone teams on vertical slices, gameplay debugging, and software architecture.

Coding Tutor — University of California, Santa Cruz 2021–2023

- Tutored Python and C++ fundamentals, object-oriented programming, and algorithmic problem solving.